

SQL Server Network & Instance Issues — Troubleshooting Guide

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Gather Information to Troubleshoot SQL Server Connection Errors

Before you start troubleshooting, gather the information listed in this section. You can use one of the following options to collect this information.

Use the SQL Check Tool to Gather the Required Information

If you can sign in locally to the SQL Server computer and have administrator access, use the SQLCHECK tool. This tool provides most of the information required for troubleshooting in one file. Review the tool's home page for more information about using the tool and the information it gathers. You can also check the recommended prerequisites and checklist page.

Collect the Data Individually Using the Following Procedures

Get the Instance Name from Configuration Manager

On the server that hosts the SQL Server instance, use SQL Server Configuration Manager to verify the instance name:

Note: Configuration Manager is automatically installed on the computer when SQL Server is installed. Instructions on starting Configuration Manager vary slightly by versions of SQL Server and Windows. For version-specific details, see SQL Server Configuration Manager.

1. Sign in to the computer hosting the instance of SQL Server.
2. Start SQL Server Configuration Manager.
3. In the left pane, select SQL Server Services.
4. In the right pane, verify the name of the instance of the database engine. • SQL SERVER (MSSQLSERVER) indicates a default instance of SQL Server. The name of the default instance is . • SQL SERVER () indicates a named instance of SQL Server. The name of the named instance is \.

Verify the Server Name in the Connection String

You often encounter errors when you specify an incorrect server name in the connection string. Make sure that the server name matches the one that you retrieved in the previous

steps.

Note: If you're using the SQLCHECK tool, review the NetBios Name/FQDN values in the Computer Information section of the output file.

For connection string references:

- For examples of connection strings, see [SQL Server Connection Strings](#).
- For more detailed examples, see [Proof of concept connecting to SQL using ADO.NET](#) under Homepage for SQL client programming.

Verify the Aliases on the Client Machines

Aliases are often used in client environments when you connect to SQL Server with an alternate name or when there are name resolution issues in the network. They're created by using SQL Server Configuration Manager or client network utility. An incorrect alias can cause the connections from your applications to connect to the wrong server, resulting in failure. Use the following methods to check for incorrect aliases. You can also use a tool such as SQLCHECK on the client machine to check for aliases and various other connectivity-related settings on a client machine.

Note: The following options only apply to the applications that use SQL Server Native Client to connect to SQL Server.

Use the SQLCHECK Output File

Test Remote Connection

Once you can connect by using TCP on the same computer, try to connect from the client computer. You can use any client application, but to avoid complexity, install the SQL Server Management tools on the client. After installation, try to use SQL Server Management Studio.

If you can connect by using shared memory, test connecting by using TCP. You can force a TCP connection by specifying tcp: before the name. Examples:

Connecting to:	Type:	Example:
Default instance	tcp:<computer name>	tcp:ACCNT27
Named Instance	tcp:<computer name/instance name>	tcp:ACCNT27\PAYROLL

7. If you can connect by using shared memory but not TCP, fix the TCP problem. The most likely issue is that TCP isn't enabled. To enable TCP, see [Verify the enabled protocols on SQL Server](#).

8. If your goal is to connect by using an account other than an administrator account, begin by connecting as an administrator. Then, try to connect again with the Windows Authentication sign-in or the SQL Server Authentication sign-in that the client application uses.

Login Failure Troubleshooting

If you receive error 18456 Login failed for user, the article MSSQLSERVER_18456 contains additional information about error codes. You can view the error log by using SSMS (if you can connect), in the Management section of Object Explorer. Otherwise, you can view the error log by using Notepad. The default location varies with your version and can be changed during setup. The default location for SQL Server 2019 (15.x) is C:\Program Files\Microsoft SQL Server\MSSQL15.MSSQLSERVER\MSSQL\Log\ERRORLOG.

Related Documentation

- Network Protocols and Network Libraries
- Client Network Configuration
- Configure Client Protocols
- Troubleshoot connectivity issues in SQL Server
- 0200 Consistent Network Issue
- Connect to an Always On availability group listener
- Always On Failover Cluster Instances (SQL Server)
- Troubleshooting connectivity issues and other errors with Azure SQL Database and Azure SQL Managed Instance

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